Quadratic Equations and Expressions

Worksheet 1 (Non-Calculator)

1. S15/qp12/q18

a. Factorise completely $p^2q - pq$ [1]

b.

i. Factorise $5x^2 + x - 4$ [1]

ii. Hence solve $5x^2 + x - 4 = 0$. [1]

2. W15/qp11/q22

a. Expand and simplify $10 - 3(3x - 2)$. [1]

b. Simplify fully $\frac{3x^2+16x+5}{9x^2-1}$. [3]

3. W15/qp12/q16(a)

a. Factorise

i. $4p^2 - 9q^2$ [1]

ii. $2n^2 + 5n - 3$. [1]

b. Express $\frac{3}{4x} + \frac{2}{3y}$ as a single fraction. [1]

4. S16/qp12/q26(b)

Simplify fully $\frac{4x^2-1}{2x^2-9x-5}$. [3]

5. S20/qp11/q7

Factorise.

a. $6c^3 + 9c$ [1]

b. $5ay - 2bx - 2by + 5ax$. [2]

6. W20/qp11/q2

a. Factorise $4p^2 - 1$. [1]

b. Factorise $10xy - 12 + 15x - 8y$. [2]

7. W20/qp12/q3

Factorise.

a. $12t^2 - 4t$. [1]

b. $a(x-y) + b(y-x)$. [1]

c. $x^2 - 2x - 3$. [1]

8. W20/qp12/q9

a. Simplify $3(3a-4) + 2(2-a)$. [2]

b. Given that $4x = 3y$, find the numerical value of $\frac{8x+y}{y}$. [1]

9. S21/qp11/q8

a. Simplify $6x + 15 - 2x + 8$. [1]

b. Expand and simplify $(x-5)^2$. [2]

10. S21/qp11/q21

Factorise.

a. $3cx + 2bx - 6cy - 4by$. [2]

b. $6x^2 + 7x - 10$. [2]

Answer Key:

1.(a)pq(p-1) (b)(i)(5x-4)(x+1) (ii)0.8, oe, 1

2.(a) 16-9x (b) $\frac{x+5}{3x-1}$

3.(i) (2p-3q)(2p+3q) (ii)(2n-1)(n+3) (b) $\frac{9y+8x}{12xy}$

4.$\frac{2x-1}{x-5}$

5. $3c(c^2+3)$ (b)(5a-2b)(x+y)

6. (a)(2p-1)(2p+1) (b)(5x-4)(2y+3)

7.(a)4t(3t-1) (b) (a-b)(x-b) (c) (x-3)(x+1)

8.(a) 7a-8 (b) 7

9.(a)4x+23 (b) $x^2 - 10x + 25$

10. (a)(3c+2b)(x-2y) (b) (6x-5)(x+2)